

Réunion d'avancement n° 14

Algorithmes de navigation par champ magnétique terrestre
pour des véhicules aérospatiaux

17/04/2025

- 3 formulations du RBPF
 - RBPF « lissant » => formulation d'Audrey
 - RBPF « à pré-correction » => formulation classique de Gustafsson
 - RBPF « recombiné » => nouvelle formulation de Hendebay
- 2 méthodes de SLAM
 - SLAM « approximation dans un espace de Hilbert »
 - SLAM « interpolation multi-capteurs »

3 formulations du RBPF

- Lissant :

$$\mathbf{x}_k := \begin{bmatrix} x_k^l \\ x_{k-1}^l \end{bmatrix} = \underbrace{\begin{bmatrix} \overline{F_k^l(x_{k-1}^n)} & 0 \\ I & 0 \end{bmatrix}}_{\mathbf{F}_k(x_{k-1}^n) :=} \begin{bmatrix} x_{k-1}^l \\ x_{k-2}^l \end{bmatrix} + \underbrace{\begin{bmatrix} Q_k^{ln} (Q_k^n)^{-1} x_k^n + \overline{f_k^l(x_{k-1}^n)} + \overline{b_k^{le}(x_{k-1}^n)} \\ 0 \end{bmatrix}}_{\mathbf{b}_k^l(x_k^n, x_{k-1}^n) :=} + \underbrace{\begin{bmatrix} \overline{w_k^l} \\ 0 \end{bmatrix}}_{\mathbf{w}_k :=}$$

$$\mathbf{z}_k := \begin{bmatrix} x_k^n \\ z_k \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & F_k^n(x_{k-1}^n) \\ H_k(x_k^n) & 0 \end{bmatrix}}_{\mathbf{H}_k(x_k^n, x_{k-1}^n) :=} \begin{bmatrix} x_k^l \\ x_{k-1}^l \end{bmatrix} + \underbrace{\begin{bmatrix} f_k^n(x_{k-1}^n) + b_k^{ne} \\ h_k(x_k^n) + b_k^o \end{bmatrix}}_{\mathbf{b}_k^o(x_k^n, x_{k-1}^n) :=} + \underbrace{\begin{bmatrix} w_k^n \\ n_k \end{bmatrix}}_{\mathbf{n}_k :=}$$

- Pré-correction :

$$(S_1) : \begin{cases} x_{k-1}^{l*} &= x_{k-1}^l \\ x_k^n &= F_k^n(x_{k-1}^n) x_{k-1}^{l*} + f_k^n(x_{k-1}^n) + b_k^{ne} + w_k^n \end{cases}$$

$$(S_2) : \begin{cases} x_k^l &= \overline{F_k^l(x_{k-1}^n)} x_{k-1}^{l*} + \overline{f_k^l(x_{k-1}^n)} + \overline{b_k^{le}} + Q_k^{nl} (Q_k^n)^{-1} x_k^n + \overline{w_k^l} \\ z_k &= H_k(x_k^n) x_k^l + h_k(x_k^n) + b_k^o + n_k \end{cases}$$

- Recombiné

$$\mathbf{x}_k := \begin{bmatrix} x_k^l \\ x_k^n \end{bmatrix} = \underbrace{\begin{bmatrix} \overline{F_k^l(x_{k-1}^n)} & 0 \\ \overline{F_k^n(x_{k-1}^n)} & 0 \end{bmatrix}}_{\mathbf{F}_k(x_{k-1}^n) :=} \begin{bmatrix} x_{k-1}^l \\ x_{k-1}^n \end{bmatrix} + \underbrace{\begin{bmatrix} \overline{f_k^l(x_{k-1}^n)} + \overline{b_k^{le}(x_{k-1}^n)} \\ f_k^n(x_{k-1}^n) + b_k^{ne} \end{bmatrix}}_{\mathbf{b}_k^c(x_k^n, x_{k-1}^n) :=} + \underbrace{\begin{bmatrix} \overline{w_k^l} \\ w_k^n \end{bmatrix}}_{\mathbf{w}_k :=}$$

$$z_k = \underbrace{[H_k(x_k^n) \ 0]}_{\mathbf{H}_k(x_k^n) :=} \begin{bmatrix} x_k^l \\ x_k^n \end{bmatrix} + \underbrace{[h_k(x_k^n) + b_k^o]}_{\mathbf{b}_k^n(x_k^n) :=} + n_k$$

RBPF lissant

$$\begin{aligned}
 \begin{bmatrix} P_{k|k-1} & P_{k-1,k|k-1} \\ (P_{k-1,k|k-1})^T & P_{k-1|k-1} \end{bmatrix} &= \mathbf{P}_{k|k-1} = \mathbf{F}_k(\hat{x}_{k-1|k-1}^n) \mathbf{P}_{k-1|k-1} \mathbf{F}_k(\hat{x}_{k-1|k-1}^n)^T + \mathbf{Q}_k \\
 &= \begin{bmatrix} \tilde{F}_k^l & 0 \\ I & 0 \end{bmatrix} \begin{bmatrix} P_{k-1|k-1} & P_{k-2,k-1|k-1} \\ P_{k-2,k-1|k-1}^T & P_{k-2|k-1} \end{bmatrix} \begin{bmatrix} (\tilde{F}_k^l)^T & I \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} \overline{Q}_k^l & 0 \\ 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} \tilde{F}_k^l P_{k-1|k-1} (\tilde{F}_k^l)^T + \overline{Q}_k^l & P_{k-1|k-1} (\tilde{F}_k^l)^T \\ \tilde{F}_k^l P_{k-1|k-1} & P_{k-1|k-1} \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \begin{bmatrix} S_k^x & S_k^{xz} \\ (S_k^{xz})^T & S_k^z \end{bmatrix} &= \mathbf{S}_k = \mathbf{H}_k(\hat{x}_{k|k}^n, \hat{x}_{k-1|k-1}^n) \mathbf{P}_{k|k-1} \mathbf{H}_k(\hat{x}_{k|k}^n, \hat{x}_{k-1|k-1}^n)^T + \mathbf{R}_k \\
 &= \begin{bmatrix} 0 & \tilde{F}_k^n \\ \tilde{H}_k & 0 \end{bmatrix} \begin{bmatrix} P_{k|k-1} & P_{k-1,k|k-1} \\ P_{k-1,k|k-1}^T & P_{k-1|k-1} \end{bmatrix} \begin{bmatrix} 0 & \tilde{H}_k^T \\ (\tilde{F}_k^n)^T & 0 \end{bmatrix} + \begin{bmatrix} Q_k^n & 0 \\ 0 & R_k \end{bmatrix} \\
 &= \begin{bmatrix} \tilde{F}_k^n P_{k-1|k-1} (\tilde{F}_k^n)^T + Q_k^n & \tilde{F}_k^n P_{k-1,k|k-1} \tilde{H}_k^T \\ \tilde{H}_k P_{k-1,k|k-1} (\tilde{F}_k^n)^T & \tilde{H}_k P_{k|k-1} \tilde{H}_k^T + R_k \end{bmatrix}
 \end{aligned}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + P_{k|k-1} \begin{bmatrix} -\tilde{H}_k^T & \tilde{H}_k^T ((S_k^z)^{-1} + (S_k^x)^{-1} (S_k^{xz})^T) \\ (\tilde{F}_k^n)^T ((S_k^x)^{-1} + (S_k^z)^{-1} S_k^{xz}) & -(\tilde{F}_k^n)^T \end{bmatrix} \begin{bmatrix} [S_k^z - (S_k^{xz})^T (S_k^x)^{-1} S_k^{xz}]^{-1} (S_k^{xz})^T (S_k^x)^{-1} x_k^n \\ [S_k^x - S_k^{xz} (S_k^z)^{-1} (S_k^{xz})^T]^{-1} S_k^{xz} (S_k^z)^{-1} z_k \end{bmatrix}$$

RBPF à pré-correction

•

$$\left(\left\{ \begin{bmatrix} x_k^{n(i)} \\ x_k^{l(i)} \end{bmatrix}, w_k^{(i)}, P_k^{l(i)} \right\}_{i=1}^N \right) = RBPF \left(\left\{ \begin{bmatrix} x_{k-1}^{n(i)} \\ x_{k-1}^{l(i)} \end{bmatrix}, w_{k-1}^{(i)}, P_{k-1}^{l(i)} \right\}_{i=1}^N, z_k \right)$$

- POUR $i = 1 : N$
 - propagation particulaire :
 - pré-correction de Kalman :

$$\begin{cases} K_k^{(i)} = P_{k-1}^{l(i)} (\hat{F}_k^n)^T (\hat{F}_k^n P_{k-1}^{l(i)} (\hat{F}_k^n)^T + Q_k^n)^{-1} \\ x_{k-1}^{l* (i)} = x_{k-1}^{l(i)} + K_k^{(i)} (x_k^{n(i)} - (\hat{F}_k^n x_{k-1}^{l(i)} + \tilde{b}_k^n)) \\ P_{k-1}^{l* (i)} = (I - K_k^{(i)} \hat{F}_k^n) P_{k-1}^{l(i)} \end{cases}$$

- prédiction de Kalman :

$$\begin{cases} x_{k|k-1}^{l(i)} = \hat{F}_k^l x_{k-1}^{l* (i)} + \tilde{b}_k^l \\ P_{k|k-1}^{l(i)} = \hat{F}_k^l P_{k-1}^{l* (i)} (\hat{F}_k^l)^T + \overline{Q}_k^l \end{cases}$$

- correction particulaire :
- correction de Kalman :

$$\begin{cases} L_k^{(i)} = P_{k|k-1}^{l(i)} \hat{H}_k^T (\hat{H}_k P_{k|k-1}^{l(i)} \hat{H}_k^T + R_k)^{-1} \\ x_k^{l(i)} = x_{k|k-1}^{l(i)} + L_k^{(i)} (z_k - (\hat{H}_k x_{k|k-1}^{l(i)} + \tilde{b}_k^o)) \\ P_k^{l(i)} = (I - L_k^{(i)} \hat{H}_k) P_{k|k-1}^{l(i)} \end{cases}$$

- FIN POUR
- POUR $i = 1 : N$
 - normalisation :
- FIN POUR

RBPF recombiné

- G. Hendeby, R. Karlsson and F. Gustafsson, "A New Formulation of the Rao-Blackwellized Particle Filter," 2007 IEEE/SP 14th Workshop on Statistical Signal Processing, Madison, WI, USA, 2007, pp. 84-88, doi: 10.1109/SSP.2007.4301223.

Algorithm 2 Rao-Blackwellized PF (New formulation)

- 1: Initialization (same as standard formulation): For $i = 1, \dots, N$, $x_{0|-1}^{p,(i)} \sim p_{x_0^p}(x_0^p)$ and set $\{\hat{x}_{0|-1}^{k,(i)}, \bar{P}_{0|-1}^{(i)}\} = \left\{ \begin{pmatrix} x_0^p \\ x_0^k \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & \bar{P}_0 \end{pmatrix} \right\}$.
Let $t = 0$. $(^{(i)})$ is used to denote particle number, and $\hat{\xi}$ and $\hat{x}^p$ indicate estimates of the parts of $\hat{x}^k$.
- 2: Measurement update

- (a) PF measurement update using (5): For $i = 1, \dots, N$, evaluate and normalize the importance weights using the likelihood

$$\begin{aligned} p(y_t | \mathbb{X}_t^p, \mathbb{Y}_{t-1}) &= \mathcal{N}(\hat{y}_t^{(i)}, S_t) \\ \hat{y}_t^{(i)} &= \bar{H}_t^{k(i)} \hat{x}_{t|t-1}^{k(i)} + u_t^{h(i)} \\ S_t^{(i)} &= \bar{H}_t^{k(i)} \bar{P}_{t|t-1}^{k(i)} (\bar{H}_{t|t-1}^{k(i)})^T + R_t, \end{aligned}$$

with $u_t^h = h(x_t^{p(i)})$.

- (b) KF measurement update using (4) (depends on the PF):

$$\begin{aligned} \hat{x}_{t|t}^{k(i)} &= \hat{x}_{t|t-1}^{k(i)} - P_t^{k(i)} (\bar{H}_{t|t-1}^{k(i)})^T (S_t^{(i)})^{-1} \epsilon_t^{(i)} \\ \epsilon_t^{(i)} &= y_t - \hat{y}_t^{(i)}, \\ \bar{P}_{t|t}^{k(i)} &= \bar{P}_{t|t-1}^{k(i)} - \bar{P}_{t|t-1}^{k(i)} (\bar{H}_{t|t-1}^{k(i)})^T (S_t^{(i)})^{-1} \bar{H}_{t|t-1}^{k(i)} \bar{P}_{t|t-1}^{k(i)}. \end{aligned}$$

- (c) Resample if necessary.

- 3: Time update

- (a) KF time update using (4):

$$\begin{aligned} \hat{x}_{t+1|t}^{k(i)} &= \bar{F}_t^{k(i)} \hat{x}_{t|t}^{k(i)} + \bar{u}_t^{k(i)} \\ \bar{P}_{t+1|t}^{k(i)} &= \bar{F}_t^{k(i)} \bar{P}_{t|t}^{k(i)} (\bar{F}_t^{k(i)})^T + \bar{G}_t^{k(i)} \bar{Q}_t^k (\bar{G}_t^{k(i)})^T \end{aligned}$$

with $\bar{u}_t^{k(i)} = (f^p(x_t^{p(i)})^T f^k(x_t^{p(i)})^T)^T$.

- (b) PF time update using (5) (depends on the KF):

$$x_{t+1}^{p(i)} \sim \mathcal{N}(\hat{\xi}_{t+1|t}^{(i)}, P_{t+1|t}^{\xi(i)})$$

- (c) Mixing step, update the KF using (6).

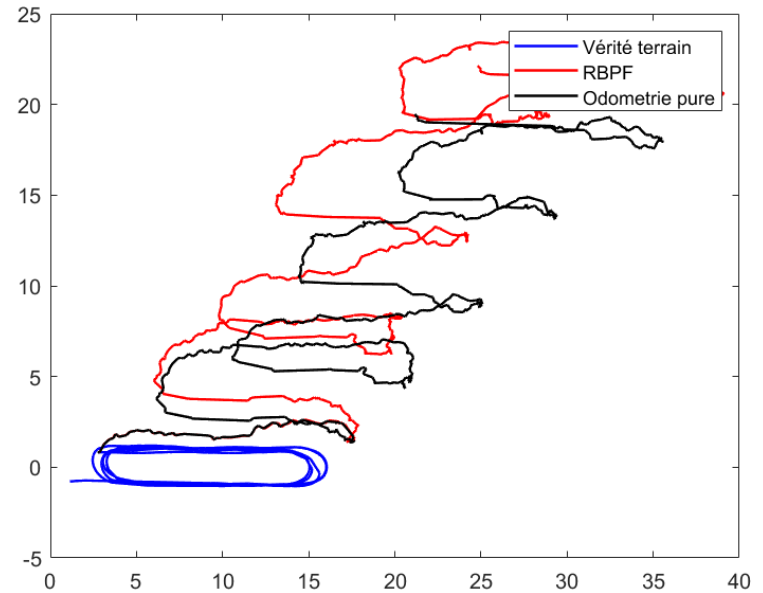
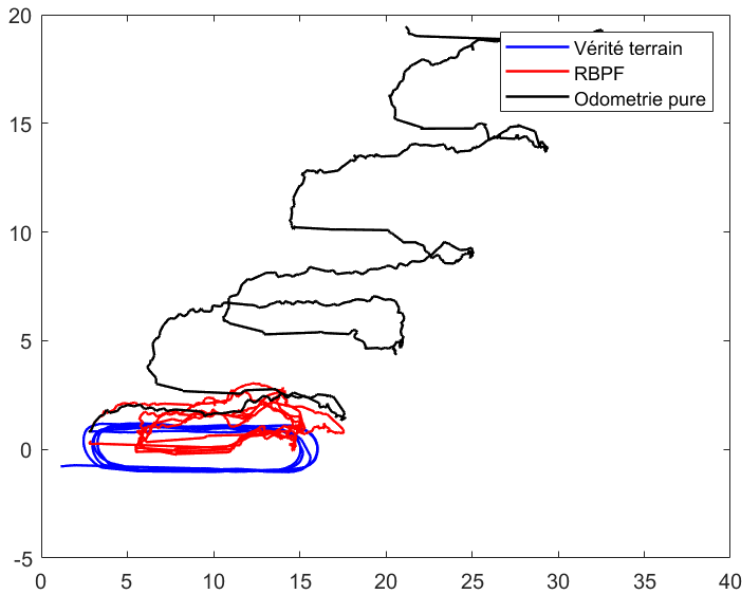
$$\begin{aligned} \hat{x}_{t+1|t}^{k(i)} &:= \hat{x}_{t+1|t}^{k(i)} + P_{t+1|t}^{k\xi(i)} (P_{t+1|t}^{\xi(i)})^{-1} (x_{t+1}^{p(i)} - \hat{\xi}_{t+1|t}^{(i)}) \\ P_{t+1|t}^{k(i)} &:= P_{t+1|t}^{k(i)} - P_{t+1|t}^{k\xi(i)} (P_{t+1|t}^{\xi(i)})^{-1} P_{t+1|t}^{\xi k(i)}. \end{aligned}$$

- 4: Increase time and repeat from step 2.
-

SLAM par approximation dans un espace de Hilbert

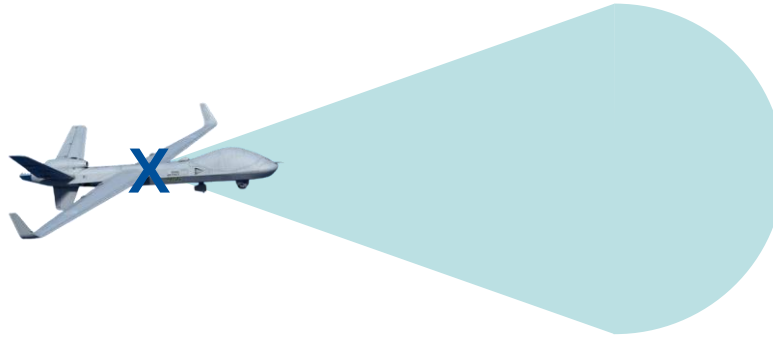
$$K(x, x') = \sum_{i=1}^{+\infty} \lambda_i \phi_i(x) \phi_i(x') \quad \Rightarrow \quad h(x) = \sum_{i=1}^{+\infty} m_i \phi_i(x) = \Phi(x)^T M, \quad m_i \sim \mathcal{N}(0, \lambda_i)$$

$$x_k = [p_x \quad p_y \quad v_x \quad v_y \quad \dots \mid m_1 \quad \dots \quad m_n]$$



SLAM multi-capteurs

- SLAM vision usuel :



- SLAM multi-capteurs :

